Current Clarity Stage 15: Final Product Semantics

Diagnostic only.

- The frozen engine is unchanged: C-gbm is not refitted, and every Stage 14 engine file matches its hash.
- Production is unchanged, and nothing was deployed.
- No Sector field observation was used, and Michael's Guntersville survey was not touched.
- Water Temperature was not started.

This is a re-score, not a blind certification.

- The Stage 14 certification set was already unsealed and scored in Stage 14, and its findings motivated this stage.
- Every new threshold here comes from development rows only, and was frozen before the re-score.
- The readiness decision requires the development rows to pass as well, so the re-score cannot carry it alone.

Step	UTC	What was fixed
Pre-registration	2026-10-01T17:04:40Z	Derivation rules for every new state, display and authority; the operational rubric; the readiness rule (`PREREGISTRATION_STAGE15.md`, `8265b7a8…`)
Semantics	2026-10-01T17:08:04Z	Parameters derived from development rows only (`engine/semantics15.json`, `a5d6da8d…`), before the re-score
Inputs	re-checked	Engine, certification sample, ACOLITE results and truth: all match their Stage 14 hashes

The answer

The semantics make the product more useful, but not yet safe enough to freeze.

- **Clear-to-Normal works.** On development it raised the operational score from 0.667 to 0.690 and cut harmful answers from 21.8% to 17.3%, so it was adopted. On the re-score its answers are right 90% of the time, and Normal moves from NOT to PARTIAL.
- **The very-clear floor did not trigger.** Predictions above 10–15 ft are as precise as any others in log terms. The deep-water error is water being under-read, and a floor on the prediction cannot reach it.
- **Algal water keeps its state.** Flagged water is still classified right 77–79% of the time on development, so the frozen rule is a warning modifier, not softening or withholding.
- **Sentinel-2C becomes lower authority.** That costs 4% of the certification rows, but 34% of the 2025-dated ones.

The new operational rubric:

- **Score:** 7/10 on both development (0.688) and the re-score (0.690).
- **Readiness to freeze is NO**, by the frozen rule, for two reasons:
  - harmful wrong answers are 15.6% on the re-score and 16.6% on development, against ≤ 15%;
  - stained water scores 0.598 and 0.563, against ≥ 0.60.

Everything else passes.

1. The product the frozen engine now gives

State	Prediction range (ft)	Display	Authority
Clear	≥ 6.94	exact feet + two-thirds interval	full
Clear-to-Normal (new)	5.00 to 6.94	exact feet + interval	full
Normal	3.80 to 5.00	exact feet + interval	full
Normal-to-Stained	2.63 to 3.80	exact feet + interval	full
Stained	2.38 to 2.63	exact feet + interval	full
Stained-to-Limited	1.70 to 2.38	exact feet + interval	full
Limited	< 1.70	exact feet + interval	full
Any state, NDCI > 0.03	as above	"~X ft" + "Algal water detected — elevated uncertainty"	full
Any state, Sentinel-2C	as above	state only (feet and interval withheld)	**lower**
Tier 3 Clear (Sen2Cor-C3 ≥ 6 ft)	—	"~X ft" (lower confidence)	**lower**
Unknown / Low Authority	any other tier-3 reading, or none	nothing	—

Exact feet are still authorised for every class. The numeric prediction behind every row is the Stage 14 prediction, unchanged.

2. Clear-to-Normal

Clear-to-Normal	Value
Band (pooled development, 2:1 odds)	5.00–6.94 ft (logistic α 0.08, β 9.7, 823 rows within ±0.30)
Nested bands, 5 folds	5.07–7.28; 4.90–6.96; 5.01–6.97; 4.87–6.99; 4.98–7.12 ft
Adoption test (development, nested)	operational 0.667 → **0.690**; harmful 21.8% → **17.3%** → **adopted**
Re-score: Clear-to-Normal answers	91, right 90% (truth: 51 clear, 31 normal, 8 stained, 1 limited)
… what Stage 14 had said for them	Clear 39, Normal 52
Plain "Normal" answers	Stage 14: 127, right 41% → Stage 15: 75, right 45%

- **How the band was set.** Exactly as for the other two: within 5.00–6.94 ft, neither side of 6 ft is twice as likely as the other on development predictions.
- **What it fixes.** It replaces 52 of the 127 Stage 14 "Normal" answers and 39 "Clear" answers. Most of them (51 of 91) were clear water read low.
- **What it leaves.** The plain "Normal" answer is now confined to 3.80–5.00 ft. It is still the weakest state (45% right), but it is issued 40% less often.

3. The very-clear floor: not adopted

Predicted ≥	Development rows	Median |log| (trigger > 0.15)	Truth – prediction (median)	Truth ≥ that depth	Truth ≥ 6 ft
10 ft	343	0.130	-0.62 ft	72%	92%
12 ft	251	0.135	-0.23 ft	68%	94%
15 ft	114	0.129	+0.19 ft	70%	94%

- **The frozen trigger was not met.** It asked whether exact feet get worse above 10, 12 or 15 ft, measured among rows predicted that deep. They do not: median |log| is 0.13 at every threshold, against the 0.15 trigger, and

predictions there are calibrated (truth − prediction −0.6 to +0.2 ft).

- **The −2.3 ft clear bias comes from somewhere else.** Very clear water is predicted too low. Of the development rows ≥ 20 ft, 82% are predicted at 12 ft or more (median 15.5 ft), and the rest are predicted lower still.
- **A floor would make the clear exact feet worse, not better.** A 12 ft floor would hide the well-predicted high rows and leave the under-read ones published: the clear median moves from −1.75 to −2.21 ft (development).
- **A "12+ ft" display would still be honest,** and remains an optional UX choice for Michael. 69% of predictions ≥ 12 ft are truly ≥ 12 ft on development (78% on the re-score), and 94–96% are clear. It is a display choice, not something the accuracy evidence requires.

4. Algal water

NDCI level	Development rows	State right (development)	Behaviour (frozen rule)	Re-score rows	State right (re-score)	In-situ algal share (dev / re-score)
moderate (0.03–0.10)	264	79%	**modifier**	117	74%	53% / 50%
strong (> 0.10)	501	77%	**modifier**	302	78%	55% / 72%

The detection. It is the existing satellite flag, NDCI > 0.03, on the answering reflectance.

- On development it catches 70% of in-situ algal water (chlorophyll ≥ 15 µg/L), with 54% precision.
- On the re-score it catches 84%.

The frozen rule chose "modifier" at both levels. The states issued to flagged water are right 79% (moderate) and 77% (strong) of the time on development. So the product shows:

What the bowfisher sees (example)
Normal-to-Stained · ~2.8 ft visibility
Algal water detected — elevated uncertainty

Why Stage 14's "algae NOT" stands. That verdict measures in-situ algal water against a constant answer: the engine is not *better* than always guessing "Stained-to-Limited" there. It is not wrong more often than elsewhere. The modifier tells the bowfisher what kind of water it is, without claiming more.

5. Sentinel-2C: lower authority (temporary)

Sentinel-2C	Value
Development S2C rows (tiers 1–2)	38; median signed log -0.050 (S2A/B +0.007)
Certification S2C rows	33 (4% of the US core); 23 lowered from full authority; feet withheld on 27
Share of 2025-dated certification rows	**34%** (85 rows)
Operational score of S2C rows	0.57; harmful answers 0

- **The rule.** S2C answers keep their state and lose full authority, and their exact feet and interval are withheld. No bias correction was fitted.
- **The evidence.** Development S2C rows read only 0.05 log low (n 38), but independent certification rows read 0.19 low (n 23). Neither sample is large enough to calibrate from.
- **When it lifts.** At least 60 independent S2C certification rows with |median signed log| ≤ 0.05, and decision quality within 0.05 of S2A/S2B.

- **The cost.** Small on this historical set, but S2C is about a third of current-era scenes. Until the rule lifts, a third of today's answers would carry "lower authority".

6. Tier 3 (unchanged)

On the re-score, Sen2Cor-C3 issued a lower-authority "Clear" for 59 of the 116 tier-3 rows, right 78% of the time. The other 57 were Unknown. 23 of those withheld an answer that would have been right; 34 withheld a wrong one or had nothing to say. Coverage was not traded for correctness.

7. The operational rubric (frozen before the re-score)

The question it answers: would this output help a bowfisher make the right visibility decision, without claiming more than the satellite supports?

Outcome (re-score, US core)	Score	Rows	Share
Correct single state	1.0	346	42%
Correct boundary state	0.9	211	26%
Correct very-clear floor	1.0	0	0%
Correct lower-authority answer	0.75	62	8%
Wrong lower-authority answer	0.25	20	2%
Appropriate Unknown	0.5	34	4%
Unknown that withheld a right answer	0.25	23	3%
Harmful wrong answer	0	129	16%

True class	Development n	Development score	Re-score n	Re-score score
Clear	745	0.848	393	**0.846**
Normal	381	0.690	177	**0.633**
Stained	227	0.563	99	**0.598**
Limited	229	0.652	156	**0.682**
Class-balanced → /10		0.688 → **7/10**		**0.690 → 7/10**
Harmful (bar ≤ 15%)		16.6%		**15.6%**
Severe (bar ≤ 2%)		2.0%		1.9%

Each layer's effect, on development:

Semantics (development, nested)	Operational	Harmful	Severe
Stage 14 states	0.667 (7/10)	21.8%	2.3%
+ floor	0.667 (7/10)	21.8%	2.3%
+ Clear-to-Normal	0.690 (7/10)	17.3%	2.0%
+ algal	0.690 (7/10)	17.3%	2.0%
+ S2C lower authority (final)	0.688 (7/10)	16.6%	2.0%

Where the harmful answers come from (re-score; development shows the same pattern):

- plain "Normal" (38; 21 of them truly clear);
- "Normal-to-Stained" on limited water (18);

- "Clear" on normal water (19);
- plain "Stained" (18).

The tiers differ. Tier-2 (relaxed-mask) answers are harmful 29% of the time, against 16% at tier 1.

8. The re-score, every rubric side by side

Rubric (certification set)	Stage 14 semantics	Stage 15 semantics
Exact feet (Stage 12)	7/10	**7/10**
Original hard class	4/10	**4/10** (unchanged by design)
Boundary-aware product (Stage 14 section 8)	2.5 / 7 → 4/10	**3.0 / 7 → 4/10**
Decision-balanced accuracy	0.726	**0.772**
Operational (new)	—	**7/10**

Capability (re-score)	Exact feet	Product, Stage 14 semantics	Product, Stage 15 semantics
Clear	PARTIAL (-2.2 ft, coverage 91%)	SUPPORTED (0.81 / 0.87)	**SUPPORTED (0.91 / 0.91)**
Normal	SUPPORTED (+0.0 ft, coverage 86%)	NOT (0.66 / 0.56)	**PARTIAL (0.76 / 0.71)**
Stained	SUPPORTED (+0.4 ft, coverage 91%)	PARTIAL (0.69 / 0.74)	**PARTIAL (0.69 / 0.75)**
Limited	SUPPORTED (+0.6 ft, coverage 89%)	PARTIAL (0.73 / 0.88)	**PARTIAL (0.73 / 0.89)**
Algae	PARTIAL (+0.5 ft, coverage 87%)	NOT (0.72 vs 0.74)	**NOT (0.73 vs 0.74)**
Sediment	SUPPORTED (+0.3 ft, coverage 87%)	PARTIAL (0.85 vs 0.88)	**PARTIAL (0.85 vs 0.88)**
Bottom-risk	NOT	NOT	**NOT**

Issued state (re-score)	n	Right	of which full authority
Clear	324	89%	264
Clear-to-Normal	91	90%	83
Normal	75	45%	70
Normal-to-Stained	100	74%	98
Stained	31	35%	29
Stained-to-Limited	73	90%	71
Limited	74	86%	71
Unknown / Low Authority	57	–	0

Splits:

Split (US core)	Group	n	Operational	Harmful	Decision-balanced (full authority)	Operational by class: clear · normal · stained · limited
platform	sentinel-2a	363	0.692 (7/10)	14.6%	0.771	0.88 · 0.67 · 0.61 · 0.62
platform	sentinel-2b	429	0.695 (7/10)	17.7%	0.767	0.83 · 0.62 · 0.59 · 0.73
platform	sentinel-2c	33	0.560 (6/10)	0.0%	–	0.62 · 0.50 · – · 0.56
season	cool	166	0.750 (8/10)	15.1%	0.829	0.77 · 0.75 · 0.70 · 0.78
season	warm	659	0.664 (7/10)	15.8%	0.745	0.86 · 0.60 · 0.56 · 0.64
tier	1	599	0.749 (7/10)	16.2%	0.793	0.88 · 0.68 · 0.65 · 0.78
tier	2	110	0.655 (7/10)	29.1%	0.695	0.87 · 0.75 · 0.57 · 0.44
tier	3	116	0.420 (4/10)	0.0%	–	0.67 · 0.25 · 0.30 · 0.46
source	NLA	155	0.690 (7/10)	15.5%	0.781	0.85 · 0.63 · 0.60 · 0.69
source	WQP	670	0.690 (7/10)	15.7%	0.771	0.85 · 0.63 · 0.60 · 0.68

- **Cool-season water now scores better than warm** (0.75 against 0.66). Clear-to-Normal absorbs the cool-season clear-water under-read that Stage 14 found.
- **S2C and tier 3 score lower by construction.** Their answers are lower authority, or Unknown.
- **SYKE** (Finland, out of domain) scores 0.557 (6/10) and stays out of every verdict.

9. Ready to freeze?

NO, by the frozen rule.

Criterion	Development	Re-score	Bar
Engine unchanged	yes	yes	yes
Operational capability	7/10	7/10	≥ 7/10
Every class ≥ 0.60	**no** (stained 0.563)	**no** (stained 0.598)	≥ 0.60
Harmful rate	**16.6%**	**15.6%**	≤ 15%
Severe rate	2.0%	1.9%	≤ 2%

- **How close it is.** The misses are narrow: 0.6 and 1.6 points of harmful rate, and 0.002 and 0.037 of stained score. All of it comes from the same place, plain single-class answers next to a boundary.
- **No further tuning here.** The obvious next rules, such as dropping the plain "Normal" and "Stained" states or lowering tier 2's authority, would have to be pre-registered on development data. They would then need a test on data no stage has seen. The Stage 14 set has now been used twice and can no longer certify anything.
- **What this does not change.** It does not move product readiness. That stays 2/10 as deployed, because production is unchanged.

10. Remaining blockers

1. **Harmful answers next to boundaries** (15.6% / 16.6%). Mostly plain "Normal" (3.80–5.00 ft), plain "Stained" (2.38–2.63 ft), and Clear/Normal confusions. Candidate rules (no plain single state inside a boundary zone; tier 2 at lower authority) need pre-registration and fresh data.
2. **Stained water** sits at 0.56–0.60 against the 0.60 bar. It is the retrieval-resolution limit found in Stage 13.
3. **No unseen certification data remains.** The P2 fresh pool, Michael's Guntersville survey, or a TVA/USACE records request are the only ways to test any further rule.
4. **Sentinel-2C** stays lower authority, about a third of current-era scenes, until ≥ 60 independent S2C rows pass.
5. **Michael's ship decision,** and the deep-water clear under-read, which is a retrieval limit no display rule fixes.
6. **Bottom-risk** still needs depth.

Files

Everything is in `scratchpad/stage15/`:

Path	Contents
`PREREGISTRATION_STAGE15.md`, `FREEZE_STAGE15.json`	The frozen rules and every hash
`engine/engine15.py`, `engine/semantics15.json`	The product layer over the unchanged Stage 14 engine
`analysis/lib15.py`	States, authority, display and the operational scorer
`analysis/dev15.py`, `dev15.json`, `floor_diag15.json`	The development derivation and evaluation
`analysis/cert15.py`, `cert15.json`	The re-score of the Stage 14 certification set

FROZEN ENGINE CHANGED: NO

CLEAR-TO-NORMAL STATE ADOPTED: YES (band 5.00–6.94 ft from development 2:1 odds; development operational 0.667 → 0.690 and harmful 21.8% → 17.3%; re-score: right 90% of 91 answers, normal NOT → PARTIAL)

VERY-CLEAR FLOOR: Not adopted. The frozen trigger (median |log| > 0.15 among predictions ≥ 10/12/15 ft) is not met: about 0.13 at every threshold. The deep-water error is under-reading, which a floor on the prediction cannot fix, and a 12 ft floor would worsen the clear exact feet (–1.75 → –2.21 ft). A "12+ ft" display would be honest (69% on development, 78% on the re-score) and remains an optional UX choice, not an accuracy rule

ALGAL PRODUCT BEHAVIOR: When NDCI > 0.03, the state is kept at full authority, visibility is shown as "~X ft", and "Algal water detected — elevated uncertainty" is attached. It is not softened or withheld, because flagged-water states are right 77–79% of the time on development (74–78% on the re-score)

S2C PRODUCT AUTHORITY: Lower authority. The state is shown; exact feet and the interval are withheld; there is no recalibration. It lifts once ≥ 60 independent S2C certification rows show |median signed log| ≤ 0.05. The cost is 4% of certification rows but 34% of 2025-dated rows

EXACT-FEET CERTIFIED CAPABILITY: 7/10 (re-score of the already-seen Stage 14 set; clear PARTIAL at –2.3 ft; normal, stained, limited and sediment SUPPORTED)

BOUNDARY-AWARE CERTIFIED CAPABILITY: 4/10 (3.0 of 7, up from 2.5: normal NOT → PARTIAL; decision-balanced 0.726 → 0.772; re-score, not blind)

OPERATIONAL BOWFISHING CLARITY CAPABILITY: 7/10 (re-score 0.690, development 0.688; harmful 15.6% / 16.6%, severe 1.9% / 2.0%)

CURRENT CLARITY READY TO FREEZE AND MOVE TO WATER TEMP: NO (frozen rule: harmful answers 15.6% on the re-score and 16.6% on development, against ≤ 15%; stained 0.598 / 0.563 against ≥ 0.60; every other criterion passes)

REMAINING BLOCKERS: (1) harmful answers next to boundaries, mostly plain "Normal" (3.80–5.00 ft), plain "Stained" and Clear/Normal confusions, plus 29% harmful at tier 2; candidate rules need pre-registration and unseen data; (2) stained water at the retrieval-resolution limit (0.56–0.60); (3) no unseen certification data remains: the P2 fresh pool, the Guntersville survey or a TVA/USACE request are needed; (4) Sentinel-2C at lower authority, about a third of current-era scenes, until ≥ 60 independent rows pass; (5) Michael's ship decision, and the deep-water clear under-read, which is a retrieval limit; (6) bottom-risk needs depth